

From Sequences to Styles: A Data-Driven Analysis of SHL Powerplay Efficiency

Andreas Arrestam and Emma Bertmar

Linköping University, Linköping, Sweden

Abstract. A powerplay represents one of the most consequential moments in ice hockey, yet conversion rates vary significantly across professional teams. This study uses Sportlogiq event data from Swedish Hockey League (SHL) games to characterise the elements separating goal-scoring sequences from unsuccessful possessions. Decomposing powerplays into distinct sequences using a ten-second inactivity threshold facilitates the extraction of features related to puck movement, zone entries and shot quality. Analysis reveals that goal sequences average around sixteen seconds shorter in duration and contain nearly twice as many shot attempts as non-goal counterparts, while using fewer carries. Through K-means clustering, three clear powerplay types emerge: pass-based, carry-heavy and point-oriented. The pass-based style associates with a superior conversion rate of 23.2%, more than five percentage points above the other two archetypes, providing actionable insights for coaches and analysts evaluating powerplay structures.

1 Introduction

A powerplay in ice hockey occurs when one team gains a numerical manpower advantage following an opponent’s penalty. These situations account for approximately 15% to 19% of total match time [2]. While powerplays are widely perceived as critical for offensive success, research indicates that even-strength play remains a more significant contributor to overall team success over a full season due to its higher frequency [2]. Despite this, gaining a powerplay drastically increases the scoring efficiency [1]. While almost all games feature even-strength goals, fewer than 75% include at least one powerplay goal, underscoring the high variance in special teams’ performance [2].

Despite their importance, traditional metrics often rely on aggregate conversion percentages, which may fail to fully capture a team’s actual tactical skill. The objective of this research is to decompose SHL powerplays into discrete sequences and identify which structural and stylistic features distinguish goal-scoring possessions from unsuccessful ones.

2 Background and Related Work

Quantitative analysis of puck possession has evolved significantly alongside player and puck tracking systems. Early modelling of possession-and-location state

spaces established the analytical foundations, treating the game as a semi-Markov process [4]. Research has noted that scoring chances are frequently generated during fast attacks and short possessions of under ten seconds [5].

Recent studies using high-resolution tracking data show large differences between players in their ability to carry the puck at high speeds, indicating that high-speed puck transport is a specialized skill primarily associated with specific player roles and tactical functions [3]. Furthermore, the value of individual goals can be assessed using metrics like the Goal Probability Importance Value (GPIV), which shows that while powerplay and even-strength goals have similar values in isolated game states, their total impact on season standings differs [2].

Specific powerplay research has historically focused on modelling efficiency via personnel and historical performance. Documentation regarding zone entries shows that controlled entries, such as carries and passes, consistently outperform dump-ins regarding shot generation. However, few studies combine sequence-level event analysis with stylistic clustering, an unsupervised machine learning approach increasingly recognized as the standard for identifying tactical archetypes in team sports [6]. This paper addresses the gap by combining sequence-level event analysis with stylistic clustering, where a powerplay sequence is defined as a continuous series of events with no gap longer than ten seconds.

3 Algorithms

Different algorithms and methods were tested to extract tactical structure from the event data before the procedures reported below were adopted.

3.1 Sequence Extraction

Events within each game were sorted by compiled game time and filtered to powerplay situations. A new sequence was started whenever the gap between consecutive events exceeded ten seconds or a period boundary was crossed, ensuring that sequences capture continuous offensive efforts. Sequences with fewer than three events, typically lost faceoffs or immediate clearances, were discarded. For each remaining sequence, the number of passes, carries, shot attempts and controlled zone entries, the duration, first controlled entry type and mean expected goal value of the shot attempts were computed.

3.2 Clustering Powerplay Goal Sequences

Powerplay goal sequences were grouped using K-means clustering to identify recurring structural patterns. Seven sequence-level features were extracted for each goal sequence: mean and minimum x-coordinate, standard deviation of y-coordinates, pass count, carry count, total event count and duration. Features were standardised to zero mean and unit variance before applying K-means with $k = 3$.

3.3 Team Style Clustering

Each team was described by five features: the share of controlled entries made by carrying the puck, the share made by passing it in, the share of entries through the centre corridor (coordinate $|y| \leq 16$), and the share of offensive-zone passes originating from the slot and from the point. Pass origins were assigned to one of five zones (point, slot, left circle, right circle, behind the net) based on rink coordinates. Features were standardised before applying K-means clustering with $k = 3$. Clusters were labelled by the feature with the highest z-scored mean, since carry entries dominate the raw values and would otherwise produce identical labels across all clusters.

3.4 Powerplay Conversion Rate

Powerplay conversion rate was computed for each team as the share of extracted sequences containing a goal. To quantify uncertainty, a bootstrap procedure with 1000 resamples was applied to each team's set of sequences, and the 2.5th and 97.5th percentiles of the resampled means were taken as a 95% confidence interval. The resulting per-team rates were compared against the league average to identify teams performing above or below expectation.

3.5 Pre-Shot Puck Movement

For each shot or goal taken under a powerplay, the number of passes made by the same team within the preceding ten seconds was counted. Shots were then grouped by this pass count and the mean expected goal value was computed for each group. Pass counts represented by fewer than ten shots were excluded to avoid unstable estimates.

4 Results

The following subsections present the main findings from the powerplay analysis.

4.1 Goal vs Non-Goal Sequences

Sequences ending in a goal differed from those that did not. As shown in Table 1, goal sequences averaged 66.7 seconds compared to 82.7 seconds for non-goal sequences. Goal sequences also averaged 4.1 shot attempts compared to 2.4 for non-goal sequences, while containing fewer carries (2.74 vs 4.17). The mean expected goal value per sequence was roughly three times higher in goal sequences (0.12 vs 0.04).

4.2 Powerplay Goal Sequence Clusters

The clustering of goal sequences produced three distinct structural types. The first cluster contains short, low-event sequences. The second represents sequences with moderate puck movement and duration. The third contains sequences with elevated pass counts and longer durations, see Figure 1.

Table 1. Mean sequence-level features for SHL powerplay possessions.

Sequence Type	Passes	Carries	Shots	Duration (s)	Mean xG
Non-goal sequence	20.55	4.17	2.39	82.73	0.04
Goal sequence	18.93	2.74	4.12	66.73	0.12

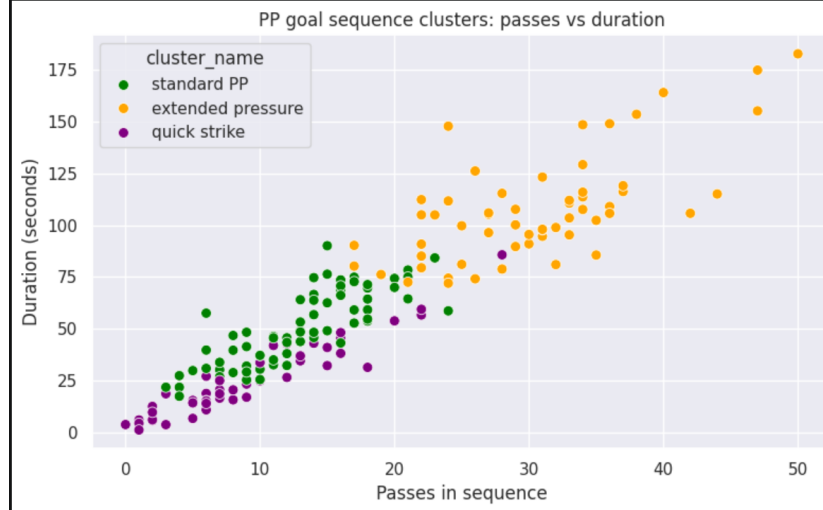


Fig. 1. Powerplay sequence based clustering using K=3

4.3 Team Style Clusters and Conversion Rate

The team-level clustering identified three powerplay archetypes. The pass-based cluster features a high frequency of pass entries and slot-oriented passing. The point-oriented cluster shows puck movement concentrated at the blue line, while the carry-heavy cluster relies on individual carries for zone entries, see Appendix Figure A1. Teams in the pass-based cluster achieved a mean conversion rate of 23.2%, compared to 17.5% for the carry-heavy cluster and 17.3% for the point-oriented cluster. Bootstrap confidence intervals at the team level showed that several teams fell clearly above or below the league average rather than within sampling noise, see Appendix Figure A2.

4.4 Powerplay Conversion Rate per Team

Powerplay conversion rates varied substantially across SHL teams, as shown in Appendix Figure A3. The league average sat around 18-19%, but individual teams ranged from below 10% to above 25%. Bootstrap confidence intervals indicate that the spread is not solely the product of sampling noise. The top and bottom teams sit clearly above and below the league average rather than overlapping with it.

5 Discussion

The results highlight rapid puck movement and high-frequency shooting as the strongest correlates of powerplay success. Goal sequences were on average sixteen seconds shorter than non-goal sequences and contained nearly twice as many shot attempts despite using fewer carries. The threefold gap in mean expected goal value (0.12 vs 0.04) indicates that they generated materially better looks at the net rather than simply benefiting from luck. The clustering of goal sequences further shows that scoring follows no single tactical formula: short low-event sequences, moderate-movement sequences and longer pass-heavy sequences all appear as recurring pathways to a goal.

The three team-level archetypes suggest a mechanism by which style translates into conversion rate. Pass-entry teams enter the zone with the puck already in motion between skaters, forcing the penalty kill to react before the attacking unit has established formation. Carry-heavy teams concede this initial movement to the carrier and require a second phase of play to generate inside ice, by which time the defensive box is typically set, while point-oriented teams move the puck along a predictable boundary that penalty kill is specifically trained to defend. The 5.7-percentage-point gap between the pass-based cluster (23.2%) and the carry-heavy cluster (17.5%) indicates that tactical philosophy acts as a major driver of efficiency, beyond raw personnel quality. Team-level conversion rates ranged from below 10% to above 25%, with non-overlapping bootstrap intervals between the top and bottom teams.

6 Conclusion

This sequence-based analysis identifies rapid puck movement and interior shot generation as the strongest correlates of powerplay success in the SHL. Goal sequences were shorter and more shot-dense than non-goal sequences, and team-level stylistic clusters explained a meaningful share of the variation in conversion rates, with the pass-based archetype outperforming the carry-heavy and point-oriented archetypes by more than five percentage points. Future work should incorporate player-level tracking data to account for individual skater positioning, and extend the dataset across multiple seasons to test the stability of the identified stylistic clusters.

References

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Code Availability

The full source code and analysis notebooks are available at <https://github.com/emmabertmar/LINHAC-project-2026>.

Appendix

This appendix presents supplementary visualisations supporting the analysis of SHL powerplay sequences. Included are the radar profiles of the identified tactical archetypes and the per-cluster conversion rates referenced in the main text.

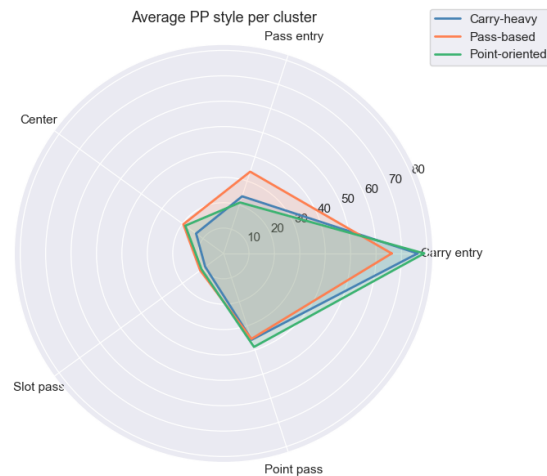


Fig. A1. Spider diagram of different team clusters

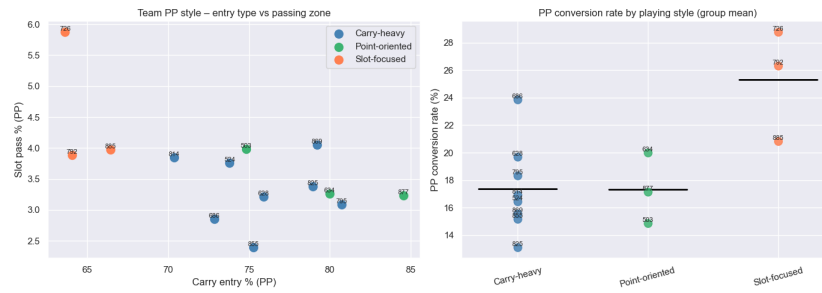


Fig. A2. Conversion rate based on team style clustering

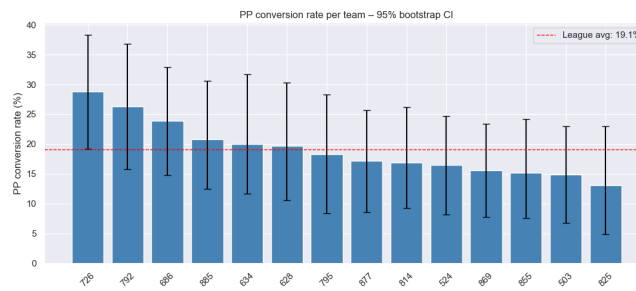


Fig. A3. Powerplay conversion rate for all teams in SHL with 95% bootstrap confidence intervals.